Foundations Notebook

Your Name

March 12, 2026

Contents

Chapter 1

Mathematical Foundations

1.1 Linear Algebra

1.1.1 Vector Spaces

Definition 1.1.1. A vector space V over a field $\mathbb{F}$ is ...

1.1.2 Why This Matters

Linear algebra underpins signal spaces, optimization, and estimation theory.

Chapter 2

Electrical Engineering Foundations

2.1 Electro Magnetism

2.1.1 Static

Electic Field

This type of electric field is based static charge, and can be thought of like gravity such that the force being exerted is all around if in a 3 d manner that diminishes with the greater range from the source.

$$F = \frac{1}{4\pi\epsilon}\left(\frac{Q_1 Q_2}{r^2}\right) \tag{2.1}$$

2.2 Radar

Radar: Radio Detection And Ranging. Is simply the transmission and receiving of Radio waves. From these radio waves it is possible to determine many things about the incident target.

Simple Model

A transmitter is to be thought as a isotropic radiator with the propogation pattern described by equation 2.2.

$$P_d = \frac{P_t}{4\pi r^2} \tag{2.2}$$

Where P_d is the power desnity at some radius, P_t is the power transmitted and r is the radius.

$$R = \frac{c\delta t}{2} \tag{2.3}$$

2.2.1 SAR

2.3 Signals and Systems

2.3.1 Adaptive Filtering

What is an Adaptive filter

An adaptive filter uses a Wiener Filter structure but iterates it and adjusts it such that the error is reduced and it begins to approach the ideal filter.

$$h(n+1) = h(n) + \Delta h(n) \tag{2.4}$$

$$\Delta h(n) = -\gamma(\frac{d\sigma^2_{e_{\min}}}{dh}) \tag{2.5}$$

Applications

Using the ides of Wiener Filters and Error Surfaces they can be applied inot the ideas of signal recovery in a noisey channel and noise cancellation.

Noise cancellation Given a the noisy signal and correlated noise we are able to filter the correlated noise such that it matches the noise of the signal and thus allows us to canel it out leaving only the signal in its place.

2.3.2 Sampling

2.3.3 LTI Systems

An LTI system satisfies:

- Linearity
- Time invariance

2.3.4 Idea

- Optical PCB's

Chapter 3

Leadership

3.1 Objective

As a military officer my job is to manage the application of violence

3.2 Culture

Between branches the ideals revolve around the unit of action. In the case of the navy the unit of action is the ship, the army is the squad, and the airforce the plane. The ability for the unit of action to execute its mission is what drives the culture. In the case of the navy, for a ship to execute its mission the entre ship crew and all must act as a unified machine. Little deviation can be accpeted as to ensure proper function of the ship. Hence the navy has a culture of detail driven instruction and execution. While the army being based around the squad needs to rely on the flexibility and inovation of the indviduals in the squad to maintian effectivness. Thus the army may focus more on the presentation and manegment of an officer as opposed to the navy's greater focus on tenichal ability.

3.3 Ideals

1. Give a shit
2. Know my shit
3. Don't be a shit
4. Get shit done